

Name:	MARKING SCHEME	Class:	Class Register Number:
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CHUNG CHENG HIGH SCHOOL (MAIN)

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**PRELIMINARY EXAMINATION 2026
SECONDARY 4**

ADDITIONAL MATHEMATICS

4049/01

Paper 1

Tuesday 25 August 2026

2 hours 15 minutes

Candidates answer on the Question Paper.

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number in the spaces at the top of this page.
 Write in dark blue or black pen.
 You may use an HB pencil for any diagrams or graphs.
 Do not use paper clips, glue or correction fluid.

Answer **all** the questions.
 Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.
 The use of an approved scientific calculator is expected, where appropriate.
 You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.
 The number of marks is given in brackets [] at the end of each question or part question.

The total number of marks for this paper is 90.

For Examiner's Use	
Question Number	Marks Obtained
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
12	
Total Marks	

This document consists of **16** printed pages and **1** blank page.

[Turn over

Mathematical Formulae**1. ALGEBRA***Quadratic Equation*

For the equation $ax^2 + bx + c = 0$,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Binomial expansion

$$(a + b)^n = a^n + \binom{n}{1} a^{n-1} b + \binom{n}{2} a^{n-2} b^2 + \dots + \binom{n}{r} a^{n-r} b^r + \dots + b^n,$$

where n is a positive integer and $\binom{n}{r} = \frac{n!}{r!(n-r)!} = \frac{n(n-1) \dots (n-r+1)}{r!}.$

2. TRIGONOMETRY*Identities*

$$\sin^2 A + \cos^2 A = 1$$

$$\sec^2 A = 1 + \tan^2 A$$

$$\operatorname{cosec}^2 A = 1 + \cot^2 A$$

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

$$\sin 2A = 2 \sin A \cos A$$

$$\cos 2A = \cos^2 A - \sin^2 A = 2 \cos^2 A - 1 = 1 - 2 \sin^2 A$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

Formulae for $\triangle ABC$

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\Delta = \frac{1}{2} ab \sin C$$

- 1 In triangle ABC , $AB = (2 + \sqrt{3})$ cm and $\angle BAC = 60^\circ$. The area of the triangle is $(6 + 3\sqrt{3})$ cm². Without using a calculator, find the length of AC , expressing your answer in the form $k\sqrt{3}$ cm, where k is an integer. [4]

Solutions

Let $AC = x$ cm,

$$\frac{1}{2}(2 + \sqrt{3})x \sin 60^\circ = 6 + 3\sqrt{3}$$

$$\frac{1}{2}(2 + \sqrt{3})x \left(\frac{\sqrt{3}}{2} \right) = 6 + 3\sqrt{3}$$

$$\frac{(2\sqrt{3} + 3)}{4}x = 6 + 3\sqrt{3}$$

$$x = \frac{4(6 + 3\sqrt{3})}{3 + 2\sqrt{3}}$$

$$x = \frac{4(6 + 3\sqrt{3})}{(3 + 2\sqrt{3})} \times \frac{(3 - 2\sqrt{3})}{(3 - 2\sqrt{3})}$$

$$= \frac{4(18 - 12\sqrt{3} + 9\sqrt{3} - 18)}{3^2 - (2\sqrt{3})^2}$$

$$= \frac{-12\sqrt{3}}{-3}$$

$$= 4\sqrt{3}$$

$$\text{B1} - \sin 60^\circ = \frac{\sqrt{3}}{2}$$

$$\text{M1} - \frac{1}{2}ab \sin C = 6 + 3\sqrt{3}$$

M1 - rationalising

A1

- 2 The points A and B lie on the curve $xy = 8$. The midpoint of AB is $M(3,3)$. Find the coordinates of A and of B . [5]

Solutions

Let $A\left(a, \frac{8}{a}\right)$ and $B\left(b, \frac{8}{b}\right)$,

M1 – coordinates for A and B

$$\frac{a+b}{2} = 3 \Rightarrow a+b = 6 \quad -(1)$$

M1 – using midpoint formula to form first equation

$$\frac{1}{2}\left(\frac{8}{a} + \frac{8}{b}\right) = 3 \Rightarrow \frac{8}{a} + \frac{8}{b} = 6$$

M1 – using midpoint formula to form second equation

$$\frac{8(a+b)}{ab} = 6 \quad -(2)$$

Sub (1) into (2): $\frac{8(6)}{ab} = 6$

M1 – simultaneous equation

$$ab = 8$$

From observation, $a = 2, b = 4$

Therefore, $A(2,4)$ and $B(4,2)$

A1

Alternative Method

Let $A\left(a, \frac{8}{a}\right)$ and $B\left(b, \frac{8}{b}\right)$,

M1 – coordinates for A and B

$$\frac{a+b}{2} = 3 \Rightarrow a+b = 6 \quad -(1)$$

M1 – using midpoint formula to form first equation

From (1), $a = 6 - b \quad -(3)$

Sub (3) into (2):

$$\frac{8}{6-b} + \frac{8}{b} = 6$$

M1 – using midpoint formula to form second equation

$$8b + 8(6-b) = 6b(6-b)$$

$$8b + 48 - 8b = 36b - 6b^2$$

$$6b^2 - 36b + 48 = 0$$

$$b^2 - 6b + 8 = 0$$

$$(b-4)(b-2) = 0$$

$$b = 4 \text{ or } b = 2$$

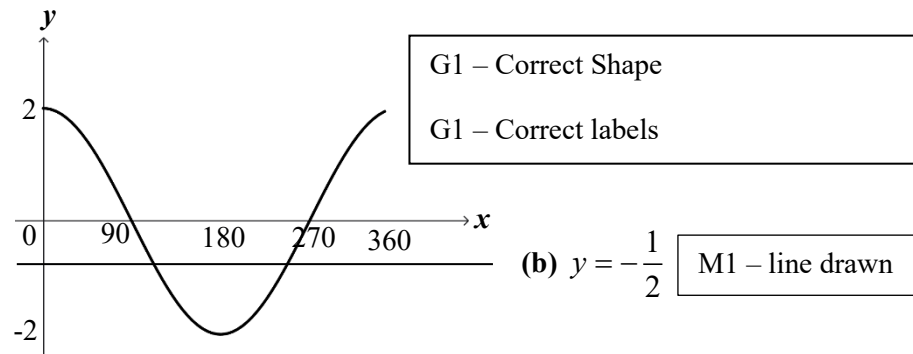
A1

Therefore, $A(2,4)$ and $B(4,2)$

- 3 (a) Sketch the graph of $y = 2 \cos x$ for $0^\circ \leq x \leq 360^\circ$.

[2]

Solutions



- (b) By adding a suitable straight line to your sketch, find the number of solutions of the equation $4 \cos x + 1 = 0$ for $0^\circ \leq x \leq 360^\circ$.

[3]

Solutions

$$4 \cos x + 1 = 0$$

$$2 \cos x = -\frac{1}{2}$$

M1 – arranging the equation to find line to be drawn

$$\therefore \text{Draw } y = -\frac{1}{2}$$

From the graph, there are 2 points of intersections for $0^\circ \leq x \leq 360^\circ$.

A1

4 A curve has equation $y = e^{2x} + 3$ and it meets the y -axis at the point P .

(a) Find the equation of the tangent to the curve at P .

[4]

Solutions

When $x = 0, y = e^0 + 3 = 4 \rightarrow P(0, 4)$ B1

$$y = e^{2x} + 3$$

$$\frac{dy}{dx} = 2e^{2x}$$

B1 – chain rule

When $x = 0, \frac{dy}{dx} = 2e^0 = 2$

Equation of tangent at $P: y - 4 = 2(x - 0)$

M1 – equation of straight line formula

$$y = 2x + 4$$

A1

The equation of the normal to the curve at P meets the x -axis at Q .

(b) Find the coordinates of Q .

[3]

Solutions

$$\text{Gradient} = -\frac{1}{2}$$

B1 – gradient of normal

Equation of normal at $P: y - 4 = -\frac{1}{2}(x - 0)$

$$y = -\frac{1}{2}x + 4$$

B1 – equation of normal

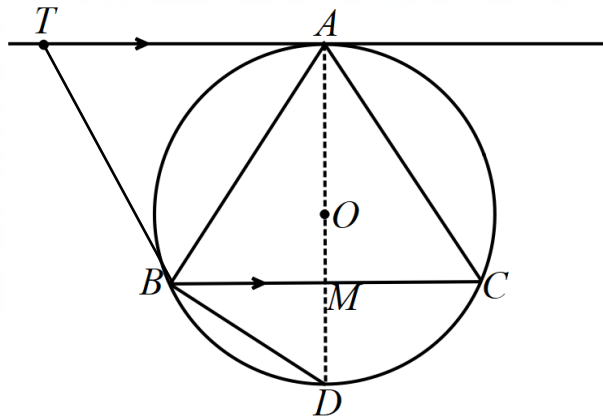
When $y = 0, -\frac{1}{2}x + 4 = 0$

$$x = 8$$

$$\therefore Q(8, 0)$$

B1

- 5 In the diagram below, TA and TB are tangents to a circle, centre O , at the points A and B respectively. The chord BC is parallel to TA and AD is the diameter of the circle.



- (a) Prove that triangle ABC is isosceles.

[3]

Solutions

$$\angle TAB = \angle ACB \text{ (tangent-chord theorem)}$$

M1 – tangent chord theorem

$$\angle TAB = \angle ABC \text{ (alternate angles, } TA \parallel BC \text{)}$$

M1 – alternate angles

Therefore, $\angle ACB = \angle ABC$, by base angles of isosceles triangle, $AB = AC$ and triangle ABC is isosceles.

A1 – with converse of base angles of isosceles triangle

- (b) Prove that triangle TAO is congruent to triangle TBO . Hence, state the name of the quadrilateral $TAOB$.

[4]

Solutions

$$\angle TAO = \angle TBO = 90^\circ \text{ (tangent } \perp \text{ radius)}$$

B1 – tangent $\perp$ radius

$$OA = OB \text{ (radii of circle)}$$

B1 – radius of circle

OT is common

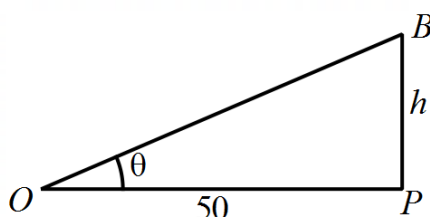
B1 – conclusion

$\therefore$ triangle TAO is congruent to triangle TBO using the RHS congruency test.

$TAOB$ is a kite.

B1

- 6 The diagram shows a balloon at B , rising vertically from a fixed point P on level ground. An observer at O is 50 m from P , and the angle of elevation of the balloon from O is θ radians. The balloon rises at a constant rate of 4 m/s.



At the instant when the balloon is 30 m above the ground, find the rate at which

- (a) the angle of elevation θ is increasing,

[5]

Solutions

$$\tan \theta = \frac{h}{50} \Rightarrow h = 50 \tan \theta \Rightarrow \frac{dh}{d\theta} = 50 \sec^2 \theta$$

$$\text{B1} - \tan \theta = \frac{h}{50}$$

M1 – differentiating h with respect to θ

$$\frac{dh}{dt} = \frac{dh}{d\theta} \times \frac{d\theta}{dt}$$

$$4 = 50 \sec^2 \theta \times \frac{d\theta}{dt}$$

M1 – forming chain rule

$$\frac{d\theta}{dt} = \frac{4}{50 \sec^2 \theta}$$

When balloon is 30 m above the ground,

$$\tan \theta = \frac{30}{50}$$

M1 – finding specific condition

$$= \frac{3}{5}$$

$$\tan^2 \theta = \frac{9}{25} \Rightarrow \sec^2 \theta - 1 = \frac{9}{25}$$

$$\sec^2 \theta = \frac{34}{25}$$

$$\frac{d\theta}{dt} = \frac{4}{50 \left(\frac{34}{25} \right)}$$

$$= \frac{4}{68}$$

A1 – conclusion

Therefore, the rate of increase of the angle of elevation is $\frac{4}{68}$ radians per second.

(b) the distance of the balloon from the observer is increasing.

[4]

Solutions

$$L^2 = 50^2 + h^2$$

B1 – Pythagoras' Theorem

$$L = \sqrt{2500 + h^2} \text{ (reject -ve)}$$

$$= (2500 + h^2)^{\frac{1}{2}}$$

$$\frac{dL}{dh} = \frac{1}{2} (2500 + h^2)^{-\frac{1}{2}} (2h)$$

M1 – differentiate L with respect to h

$$= \frac{h}{\sqrt{2500 + h^2}}$$

$$\frac{dL}{dt} = \frac{dL}{dh} \times \frac{dh}{dt}$$

$$= \frac{4h}{\sqrt{2500 + h^2}}$$

M1 – Chain Rule

When $h = 30$,

$$\begin{aligned} \frac{dL}{dt} &= \frac{120}{\sqrt{2500 + 900}} \\ &= 2.06 \end{aligned}$$

A1 – with conclusion

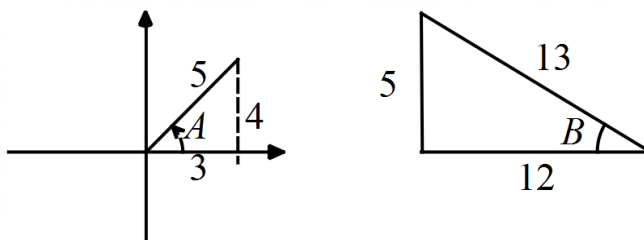
Therefore, the rate of increase of distance between observer and balloon is 2.06 m per seconds.

- 7 Given that $\cos A = \frac{3}{5}$ and $\sin B = \frac{5}{13}$, where A and B are acute and $A > B$, find the exact value of

(a) $\sin 2A$,

[2]

Solutions



$$\sin 2A = 2 \sin A \cos A$$

$$= 2 \left(\frac{4}{5} \right) \left(\frac{3}{5} \right)$$

$$= \frac{24}{25}$$

$$\text{B1 - } \sin A = \frac{4}{5}$$

B1

(b) $\cos(A + B)$,

[2]

Solutions

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

$$= \left(\frac{3}{5} \right) \left(\frac{12}{13} \right) - \left(\frac{4}{5} \right) \left(\frac{5}{13} \right)$$

$$= \frac{36}{65} - \frac{20}{65}$$

$$= \frac{16}{65}$$

$$\text{B1 - } \cos B = \frac{12}{13}$$

B1

- (c) By writing $A - B = 2A - (A + B)$ and using your results to parts (a) and (b), find the exact value of $\sec(A - B)$.

[4]

Solutions

$$\cos(A - B) = \cos 2A \cos(A + B) + \sin 2A \sin(A + B)$$

B1 – correct decomposition

$$= \left(-\frac{7}{25} \right) \left(\frac{16}{65} \right) + \left(\frac{24}{25} \right) \left(\frac{63}{65} \right)$$

$$= \frac{56}{65}$$

$$\text{B1 - } \cos 2A = -\frac{7}{25}$$

$$\text{B1 - } \sin(A + B) = \frac{63}{65}$$

$$\sec(A - B) = \frac{65}{56}$$

B1

8 A bank offers two saving schemes for a small annual interest rate x , invested over n years.

- **Scheme A:** Amount invested multiplies by $(1+x)^n$,
- **Scheme B:** Amount invested multiplies by $(1+2x)^6$.

- (a) The bank designs Scheme A so that in the expansion of $(1+x)^n$, the coefficient of x^2 is 66. Show that $n=12$. [2]

Solutions

$$\frac{n(n-1)}{2} = 66$$

M1 – equating coefficient to 66

$$n^2 - n = 132$$

$$n^2 - n - 132 = 0$$

$$(n-12)(n+11) = 0$$

$$n = 12 \text{ or } n = -11 \text{ (rejected)}$$

A1

- (b) Write down the expansions of $(1+x)^{12}$ and $(1+2x)^6$, each in ascending powers of x up to and including the term in x^2 . [2]

Solutions

$$(1+x)^{12} = 1 + 12x + 66x^2 + \dots$$

B1

$$(1+2x)^6 = 1 + \binom{6}{1}(2x) + \binom{6}{2}(2x)^2 + \dots$$

$$= 1 + 12x + 60x^2 + \dots$$

B1

- (c) Hence, determine which scheme gives the greater return for small positive values of x , justifying your answer. [2]

Solutions

Both expansions have the same constant term and the same x term. The only difference is the x^2 term. Scheme A gives the greater return by $(66-60)x^2 = 6x^2$.

B1 – Scheme A

B1 - Difference

- (d) Given that $x=0.02$, estimate the difference between the two returns and hence the difference in the final amount for an initial investment of \$20 000. [2]

Solutions

At $x=0.02$, the difference in growth factors is

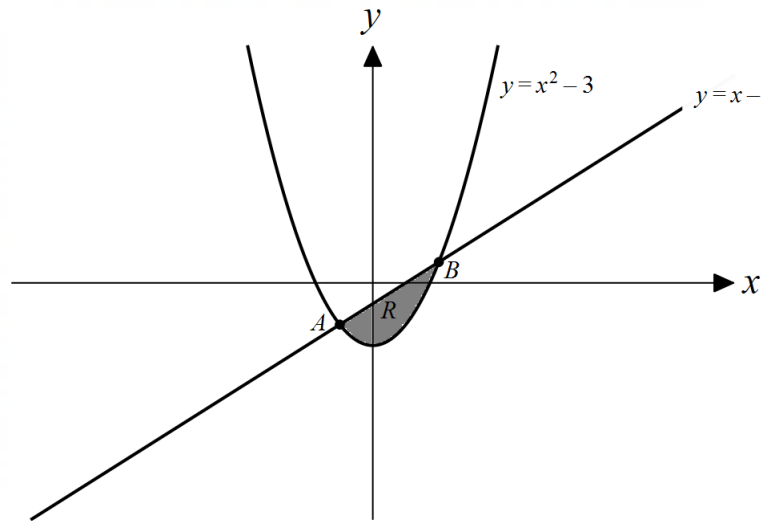
$$6x^2 = 6(0.02)^2 = 0.0024.$$

B1 – 0.0024

For an initial investment of \$20 000, the difference in the final amount = \$48.

B1 – \$48

- 9 The diagram shows the curve $y = x^2 - 3$ intersecting the line $y = x - 1$ at the points A and B . The shaded region R is enclosed by the curve and the line.



- (a) Find the coordinates of A and of B .

[4]

Solutions

$$x^2 - 3 = x - 1$$

$$x^2 - x - 2 = 0$$

$$(x - 2)(x + 1) = 0$$

$$x = -1 \text{ or } 2$$

$$\text{When } x = -1, y = -2$$

$$\text{When } x = 2, y = 1$$

$$\therefore A(-1, -2) \text{ and } B(2, 1)$$

M1 – Solving Quadratic Equation

M1 – Solving for y coordinates

A1 – for A

A1 – for B

- (b) Find the area of the shaded region R .

[4]

Solutions

$$\text{Area of shaded region } R = \int_{-1}^2 x - 1 - (x^2 - 3) \, dx$$

$$= \int_{-1}^2 -x^2 + x + 2 \, dx$$

$$= \left[-\frac{x^3}{3} + \frac{x^2}{2} + 2x \right]_{-1}^2$$

$$= \frac{10}{3} - \left(-\frac{7}{6} \right)$$

$$= \frac{9}{2} \text{ unit}^2$$

M1 – formulate integration

B1 – correct integration

M1 – upper - lower

A1

- (c) For a value of x in the interval between the x -coordinate of A and B , the vertical distance between the line and the curve is denoted by p . Express p in the form $a - (x - b)^2$, and hence state the maximum value of p . [5]

Solutions

$$p = 2 + x - x^2$$

B1

$$= -(x^2 - x) + 2$$

M1 – factorise -1

$$= -\left[x^2 - x + \left(\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2 \right] + 2$$

M1 – completing the square method

$$= -\left[\left(x - \frac{1}{2}\right)^2 - \frac{1}{4} \right] + 2$$

$$= -\left(x - \frac{1}{2}\right)^2 + \frac{1}{4} + 2$$

$$= -\left(x - \frac{1}{2}\right)^2 + \frac{9}{4}$$

A1

Hence the maximum value of p is $\frac{9}{4}$. B1

- 10 (a) Express $\frac{9x+9}{(x-1)(x+2)^2}$ in partial fractions.

[4]

Solutions

$$\frac{9x+9}{(x-1)(x+2)^2} = \frac{A}{x-1} + \frac{B}{x+2} + \frac{C}{(x+2)^2}$$

M1 – partial fraction

$$9x+9 = A(x+2)^2 + B(x-1)(x+2) + C(x-1)$$

$$\text{When } x=1, 18 = A(1+2)^2$$

$$18 = 9A$$

$$A = 2$$

M1 – finding 2 out of 3 unknown

M1 – finding all unknowns

$$\text{When } x=-2, -9 = C(-2-1)$$

$$C = 3$$

$$\text{When } x=0, 9 = 2(4) + B(-1)(2) + 3(-1)$$

$$9 = 8 - 2B - 3$$

$$4 = -2B$$

$$B = -2$$

$$\therefore \frac{9x+9}{(x-1)(x+2)^2} = \frac{2}{x-1} - \frac{2}{x+2} + \frac{3}{(x+2)^2}$$

A1

- (b) Hence, find $\int \frac{9x+9}{(x-1)(x+2)^2} dx$.

[2]

Solutions

$$\int \frac{9x+9}{(x-1)(x+2)^2} dx = \int \frac{2}{x-1} - \frac{2}{x+2} + \frac{3}{(x+2)^2} dx$$

M1 – using (a) result

$$= 2\ln(x-1) - 2\ln(x+2) - \frac{3}{x+2} + c$$

A1

- 11 (a) Express $8\sin\theta + 6\cos\theta$ in the form $R\sin(\theta + \alpha)$, where $R > 0$ and $0^\circ < \alpha < 90^\circ$. [3]

Solutions

$$R\sin(\theta + \alpha) = R\sin\theta\cos\alpha + R\cos\theta\sin\alpha$$

By comparing, $R\cos\alpha = 8$, $R\sin\alpha = 6$.

$$\begin{aligned} R &= \sqrt{8^2 + 6^2} \\ &= \sqrt{100} \\ &= 10 \end{aligned}$$

M1 – finding R

$$\tan\alpha = \frac{6}{8}$$

M1 – finding α

$$\alpha = 36.869\dots^\circ$$

$$\therefore 8\sin\theta + 6\cos\theta = 10\sin(\theta + 36.9^\circ)$$

A1

- (b) Hence, solve the equation $8\sin\theta + 6\cos\theta = 5$ for $0^\circ \leq \theta \leq 360^\circ$. [4]

Solutions

$$\text{From (a), } 10\sin(\theta + 36.869\dots^\circ) = 5$$

M1 – “hence”

$$\sin(\theta + 36.869\dots^\circ) = 0.5$$

$$\text{Basic angle} = 30^\circ$$

M1 – finding basic angle

$$\theta + 36.869\dots^\circ = 180^\circ - 30^\circ \text{ or } \theta + 36.869\dots^\circ = 360^\circ + 30^\circ$$

M1 – finding either answer

$$\theta = 113.131\dots^\circ \text{ or } \theta = 353.131\dots^\circ$$

$$\therefore \theta = 113.1^\circ \text{ or } 353.1^\circ$$

A1

- (c) State the maximum value of $4 + 8\sin\theta + 6\cos\theta$ and the value of θ , where $0^\circ \leq \theta \leq 360^\circ$, at which the maximum occurs. [2]

Solutions

$$8\sin\theta + 6\cos\theta + 4 = 10\sin(\theta + 36.869\dots^\circ) + 4$$

Since maximum value of $\sin(\theta + 36.869\dots^\circ)$ is 1, maximum value of $8\sin\theta + 6\cos\theta + 4 = 10(1) + 4 = 14$ occurring when

$$\sin(\theta + 36.869\dots^\circ) = 1$$

B1

$$\theta + 36.869\dots^\circ = 90^\circ$$

$$\theta = 53.131\dots^\circ$$

$$= 53.1^\circ$$

B1

- 12 A cuboid has a square base of side x cm and height h cm. The total length of all twelve edges is 36 cm.

(a) Show that the volume of the cuboid, $V \text{ cm}^3$, is given by $V = x^2(9 - 2x)$. [2]

Solutions

$$8x + 4h = 36$$

$$4h = 36 - 8x$$

$$h = 9 - 2x$$

$$\text{B1} - h = 9 - 2x$$

Volume = base area $\times$ height

$$= x \times x \times h$$

$$= x^2(9 - 2x)$$

B1

(b) Explain why the value of x must satisfy $0 < x < 4.5$. [2]

Solutions

B1 – length is positive

Since x is a length, it must be positive value. h is the height of the cuboid and it must also be positive. Since $h = 9 - 2x$, $9 - 2x > 0 \Rightarrow x < 4.5$.

$$\therefore 0 < x < 4.5$$

B1

(c) Given that x can vary, find the maximum volume of the cuboid and the value of x at which this occurs. Justify that this volume is a maximum. [5]

Solutions

$$V = x^2(9 - 2x)$$

$$= 9x^2 - 2x^3$$

$$\frac{dV}{dx} = 18x - 6x^2$$

M1 – differentiating V

At stationary point, $\frac{dV}{dx} = 0$

$$18x - 6x^2 = 0$$

M1 – equating to zero

$$6x(3 - x) = 0$$

$$x = 0 \text{ or } x = 3$$

Since $0 < x < 4.5$, $x = 3$.

M1 – solving $\frac{dV}{dx} = 0$

$$\text{Max } V = 3^2(9 - 2(3))$$

$$= 27 \text{ cm}^3$$

A1 – solving max V

$$\frac{d^2V}{dx^2} = 18 - 12x$$

When $x = 3$, $\frac{d^2V}{dx^2} = 18 - 12(3) = -18 < 0$

B1 – any method of proving maximum

By Second Derivative Test, the volume of the cuboid is maximum when $x = 3$.